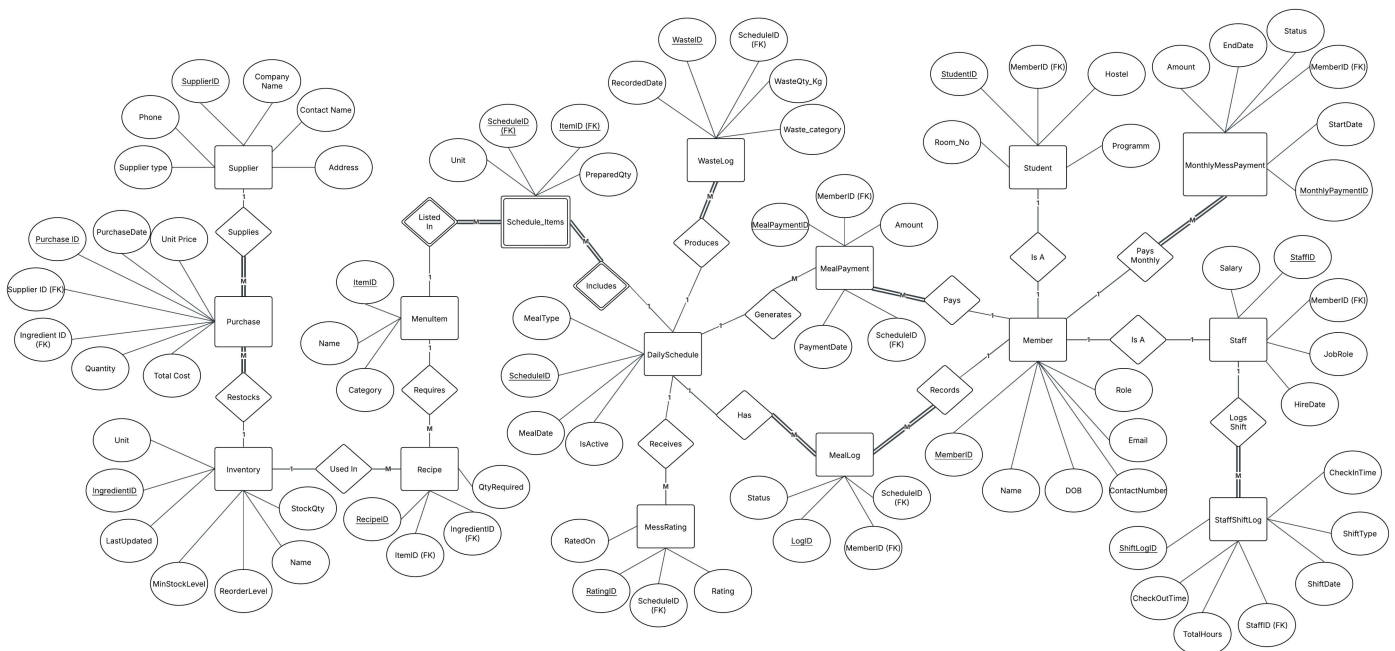


Module B: Conceptual Design (CS432 Track 1 Assignment 1)

Project Name: Mess Management System

1. ER Diagram

Note: This diagram uses **Chen's Notation** as requested. Entities are represented as **Rectangles**, and Relationships are represented as **Diamonds**. Primary Keys are underlined.



2. Explanations & Justifications

UML to ER Mapping Transition

The transition from the Object-Oriented design to the Relational ER model follows these rules:

- 1. **Inheritance Mapping:** The `Member` table contains common attributes. `Student` and `Staff` tables link to it via `MemberID` (1:1 relationship) to handle specialization.
- 2. **M:N Relationships:** The relationship between `DailySchedule` and `MenuItem` is resolved by the `Schedule_Items` table, which tracks `'QuantityPrepared'` for each item in a meal.
- 3. **Aggregation:** `Recipe` links `MenuItem` and `Inventory`, defining ingredients required for dishes.

Relationship Justifications

- **Payment Separation:** We separated `MonthlyMessPayment` (for regular fees) and `MealPayment` (for individual meal purchases) to handle different billing cycles effectively.
- **Staff Management:** `StaffShiftLog` tracks working hours precisely with check-in/out times, linked to the main `staff` entity.

Constraints & Integrity

- **Primary Keys (PK):** Uniquely identify each record. Marked clearly in the diagram.
- **Foreign Keys (FK):** Establish relationships between tables. All Foreign Keys use `'ON DELETE CASCADE'` (as seen in the SQL) to ensure data consistency.
- **Domain Checks:** Constraints like `Salary > 0` and `Status` enums are enforced via SQL `CHECK` constraints, while logical domain constraints like an Age restriction (`Age >= 16`) are derived dynamically from the `DOB` attribute.

3. Team Members & Contributions

Team Name: Interview Rejected

Name	Contribution
Sai Teja [23110366]	ER Diagram, Wrote SQL Queries.
Moksh Jindal [23110210]	Wrote SQL Queries, Designing of Database.
Mohit Yadav [23110209]	Designing of Database.
Tejas Patil [23110333]	Compiled the report.

Name	Contribution
Ankit Singh	NA